

Task 1: System Monitoring Setup

install Monitoring Tools:

```
sudo apt update && sudo apt install htop -y
```

```
aira@aira-pc: ~
* Support: https://ubuntu.com/advantage

System information as of Sun Mar 22 11:37:42 IST 2026

System load: 0.48      Processes: 66
Usage of /: 0.2% of 1006.85GB   Users logged in: 1
Memory usage: 12%      IPv4 address for eth0: 172.28.66.33
Swap usage: 0%

* Ubuntu 20.04 LTS Focal Fossa has reached its end of standard support
on 31 May 2025.

For more details see:
https://ubuntu.com/20-04

Expanded Security Maintenance for Infrastructure is not enabled.

124 updates can be applied immediately.
32 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

85 additional security updates can be applied with ESM Infra.
Learn more about enabling ESM Infra service for Ubuntu 20.04 at
https://ubuntu.com/20-04

This message is shown once a day. To disable it please create the
/home/aira/.hushlogin file.
aira@aira-pc:~$ ls
largefile.bin
aira@aira-pc:~$ sudo apt update && sudo apt install htop -y
[sudo] password for airta:
Hit:1 http://archive.ubuntu.com/ubuntu focal InRelease
Get:2 http://security.ubuntu.com/ubuntu focal-security InRelease [128 kB]
Get:3 http://archive.ubuntu.com/ubuntu focal-updates InRelease [128 kB]
Get:4 http://archive.ubuntu.com/ubuntu focal-backports InRelease [128 kB]
Fetched 383 kB in 2s (207 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
57 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree
Reading state information... Done
htop is already the newest version (2.2.0-2build1).
htop set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 57 not upgraded.
aira@aira-pc:~$
```

Command: run the htop to get to know the CPU and Memory usage
htop

```
aira@aira-pc: ~
1 [ 0.0%] 5 [ 0.0%] 9 [ 0.0%] 13 [ 0.0%]
2 [ 0.0%] 6 [ 0.0%] 10 [ 0.0%] 14 [ 0.0%]
3 [ 0.0%] 7 [ 0.0%] 11 [ 0.0%] 15 [ 0.0%]
4 [ 0.0%] 8 [ 0.0%] 12 [ 0.0%] 16 [ 0.0%]
Mem[|||||] 384M/3.68G Tasks: 35, 50 thr; 1 running
Swp[ ] 0K/1.00G Load average: 0.37 0.09 0.03
Uptime: 00:00:20

PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
737 airta 20 0 8164 3712 3072 R 0.0 0.1 0:00.01 htop
1 root 20 0 166M 10740 7924 S 0.0 0.3 0:00.64 /sbin/init
9 root 20 0 3120 2048 1920 S 0.0 0.1 0:00.00 /init
2 root 20 0 3120 2048 1920 S 0.0 0.1 0:00.00 /init
8 root 20 0 3296 1908 1792 S 0.0 0.0 0:00.00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --
7 root 20 0 3296 1908 1792 S 0.0 0.0 0:00.00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --
62 root 19 -1 43928 14848 14080 S 0.0 0.4 0:00.19 /lib/systemd/systemd-journald
84 root 20 0 23332 5632 3840 S 0.0 0.1 0:00.23 /lib/systemd/systemd-udev
96 systemd-n 20 0 19088 7296 6528 S 0.0 0.2 0:00.06 /lib/systemd/systemd-networkd
281 root 20 0 363M 7056 1280 S 0.0 0.2 0:00.05 snapfuse /var/lib/snapd/snaps/core20_1891.snap /snap/co
283 root 20 0 363M 7056 1280 S 0.0 0.2 0:00.04 snapfuse /var/lib/snapd/snaps/core20_1891.snap /snap/co
500 root 20 0 363M 7056 1280 S 0.0 0.2 0:00.04 snapfuse /var/lib/snapd/snaps/core20_1891.snap /snap/co
520 root 20 0 363M 7056 1280 S 0.0 0.2 0:00.00 snapfuse /var/lib/snapd/snaps/core20_1891.snap /snap/co
521 root 20 0 363M 7056 1280 S 0.0 0.2 0:00.00 snapfuse /var/lib/snapd/snaps/core20_1891.snap /snap/co
280 root 20 0 363M 7056 1280 S 0.0 0.2 0:00.16 snapfuse /var/lib/snapd/snaps/core20_1891.snap /snap/co
285 root 20 0 85588 1280 1152 S 0.0 0.0 0:00.00 snapfuse /var/lib/snapd/snaps/lxd_24061.snap /snap/lxd/
286 root 20 0 85588 1280 1152 S 0.0 0.0 0:00.00 snapfuse /var/lib/snapd/snaps/lxd_24061.snap /snap/lxd/
282 root 20 0 85588 1280 1152 S 0.0 0.0 0:00.00 snapfuse /var/lib/snapd/snaps/lxd_24061.snap /snap/lxd/
290 root 20 0 219M 1408 1152 S 0.0 0.0 0:00.00 snapfuse /var/lib/snapd/snaps/lxd_38333.snap /snap/lxd/
```

Command : Run the df -kh , to know the files/folder size in the path
df -kh

```
aira@aira-pc: ~  
aira@aira-pc:~$  
aira@aira-pc:~$  
aira@aira-pc:~$ df -kh  
Filesystem      Size  Used Avail Use% Mounted on  
none            1.9G   0 1.9G   0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2  
none            1.9G  4.0K 1.9G   1% /mnt/wsl  
drivers         202G  107G  96G  53% /usr/lib/wsl/drivers  
/dev/sdd        1007G   2.0G  954G   1% /  
none            1.9G   76K 1.9G   1% /mnt/wslg  
none            1.9G   0 1.9G   0% /usr/lib/wsl/lib  
rootfs          1.9G  2.7M 1.9G   1% /init  
none            1.9G   0 1.9G   0% /dev  
none            1.9G  876K 1.9G   1% /run  
none            1.9G   0 1.9G   0% /run/lock  
none            1.9G   0 1.9G   0% /run/shm  
none            1.9G  76K 1.9G   1% /mnt/wslg/versions.txt  
none            1.9G  76K 1.9G   1% /mnt/wslg/doc  
C:\             202G  107G  96G  53% /mnt/c  
D:\             275G  40G  235G  15% /mnt/d  
snapfuse        64M   64M   0 100% /snap/core20/1891  
snapfuse        92M   92M   0 100% /snap/lxd/24061  
snapfuse        92M   92M   0 100% /snap/lxd/38333  
snapfuse        54M   54M   0 100% /snap/snapd/19122  
snapfuse        49M   49M   0 100% /snap/snapd/26382  
tmpfs           377M  4.0K  377M   1% /run/user/1000  
aira@aira-pc:~$
```

Command : Run the du -sh folder_name , To get to know the which particular folder is consumed space

sudo du -sh workspace/

```
aira@aira-pc:/home/sarah$ sudo du -sh workspace/  
4.0K    workspace/  
aira@aira-pc:/home/sarah$
```

Sorting:

we can pipe the results to sort -h to see the largest items at the bottom:

du -sh /folder_name/* | sort -h

```
aira@aira-pc:/home/sarah$ sudo du -sh /home/sarah/* | sort -h  
4.0K    /home/sarah/backup.sh  
4.0K    /home/sarah/workspace  
aira@aira-pc:/home/sarah$
```

Commands : To check the CPU and the Memory Resources

sudo ps aux --sort=-%cpu | head -10

```

aira@aira-pc:/home/sarah$ sudo ps aux --sort=-%cpu | head -10
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root        296  0.2  0.3 593924 12284 ?        Ssl   21:06   0:02 snapfuse /var/lib/snapd/snaps/snapd_26382.snap /snap/
snapd/26382 -o ro,nodev,allow_other,suid
root         1  0.0  0.2 170000 10740 ?        Ss    21:06   0:00 /sbin/init
root         2  0.0  0.0   3120   2048 ?        Sl    21:06   0:00 /init
root         7  0.0  0.0   3148   1792 ?        Sl    21:06   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8
--pipe-fd 10 --log-truncate
root        62  0.0  0.3  52124 14848 ?        S<s   21:06   0:00 /lib/systemd/systemd-journald
root        84  0.0  0.1  23332  5632 ?        Ss    21:06   0:00 /lib/systemd/systemd-udev
systemd+    96  0.0  0.1  19088  7296 ?        Ss    21:06   0:00 /lib/systemd/systemd-networkd
root       280  0.0  0.1 372728  7056 ?        Ssl   21:06   0:00 snapfuse /var/lib/snapd/snaps/core20_1891.snap /snap/
core20/1891 -o ro,nodev,allow_other,suid
root       282  0.0  0.0 151124  1408 ?        Ssl   21:06   0:00 snapfuse /var/lib/snapd/snaps/lxd_24061.snap /snap/lx
d/24061 -o ro,nodev,allow_other,suid
aira@aira-pc:/home/sarah$

```

sudo ps aux --sort=-%mem | head -10

```

aira@aira-pc:/home/sarah$ sudo ps aux --sort=-%mem | head -10
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root      323  0.0  0.9 2219440 37108 ?        Ssl   21:06   0:00 /usr/lib/snapd/snapd
root      385  0.0  0.5 108136 20692 ?        Ssl   21:06   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unatt
ended-upgrade-shutdown --wait-for-signal
root      315  0.0  0.4  29880 17040 ?        Ss    21:06   0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-s
tartup-triggers
root         62  0.0  0.3  52124 14976 ?        S<s   21:06   0:00 /lib/systemd/systemd-journald
systemd+   305  0.0  0.3  25588 13696 ?        Ss    21:06   0:00 /lib/systemd/systemd-resolved
root       296  0.2  0.3 593924 12284 ?        Ssl   21:06   0:02 snapfuse /var/lib/snapd/snaps/snapd_26382.snap /snap/
snapd/26382 -o ro,nodev,allow_other,suid
root       331  0.0  0.2 393352 11392 ?        Ssl   21:06   0:00 /usr/lib/udisks2/udisksd
root       343  0.0  0.2 315112 10752 ?        Ssl   21:06   0:00 /usr/sbin/ModemManager
root         1  0.0  0.2 170000 10740 ?        Ss    21:06   0:00 /sbin/init

```

Disk & Process Monitoring Script:

Create a script `/usr/local/bin/sys_monitor.sh` to log system health:

sys_monitor.sh code:

```

#!/bin/bash
LOG_FILE="/var/log/sys_health.log"
echo "--- Report Date: $(date) ---" >> $LOG_FILE
echo "Disk Usage:" >> $LOG_FILE df -h >> $LOG_FILE
echo "Top 5 Resource Intensive Processes:" >> $LOG_FILE
ps -eo pid,ppid,cmd,%mem,%cpu --sort=-%cpu | head -n 6 >> $LOG_FILE
echo "-----" >> $LOG_FILE

```

Run `chmod +x /usr/local/bin/sys_monitor.sh` to make it executable.

```

aira@aira-pc:~$ vim sys_monitor.sh
aira@aira-pc:~$ vim sys_monitor.sh
aira@aira-pc:~$ vim sys_monitor.sh
aira@aira-pc:~$ chmod +X sys_monitor.sh
aira@aira-pc:~$

```

Task 2: User Management & Access Control

Create Users and Workspaces:

sudo useradd -m sarah

sudo useradd -m mike

```
aira@aira-pc: ~  
/etc/passwd  
aira@aira-pc:~$ cat /etc/passwd  
root:x:0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
bin:x:2:2:bin:/bin:/usr/sbin/nologin  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
games:x:5:60:games:/usr/games:/usr/sbin/nologin  
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin  
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin  
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin  
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin  
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin  
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin  
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin  
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin  
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin  
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin  
gnats:x:41:41:Gnats Bug-Reporting System (admin)/var/lib/gnats:/usr/sbin/nologin  
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin  
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin  
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin  
systemd-timesync:x:102:104:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin  
messagebus:x:103:106:/:nonexistent:/usr/sbin/nologin  
syslog:x:104:110:/:home/syslog:/usr/sbin/nologin  
_apt:x:105:65534:/:nonexistent:/usr/sbin/nologin  
tss:x:106:111:TPM software stack,,,:/var/lib/tpm:/bin/false  
uidd:x:107:112:/:run/uidd:/usr/sbin/nologin  
tcpdump:x:108:113:/:nonexistent:/usr/sbin/nologin  
sshd:x:109:65534:/:run/sshd:/usr/sbin/nologin
```

check created users:

ls -ld /home/sarah

ls -ld /home/mike

```
aira@aira-pc:~$ ls -ld /home/mike  
drwxr-xr-x 2 mike mike 4096 Mar 22 11:52 /home/mike  
aira@aira-pc:~$ ls -ld /home/sara  
ls: cannot access '/home/sara': No such file or directory  
aira@aira-pc:~$ ls -ld /home/sarah  
drwxr-xr-x 2 sarah sarah 4096 Mar 22 11:52 /home/sarah  
aira@aira-pc:~$
```

Set set passwords for Created users

echo "sarah:SecurePass123!" | sudo chpasswd

echo "mike:SecurePass456!" | sudo chpasswd

```
aira@aira-pc:~$ echo "sarah:SecurePass123!" | sudo chpasswd  
aira@aira-pc:~$ echo "mike:SecurePass456!" | sudo chpasswd  
aira@aira-pc:~$ sudo passwd -S sarah  
sarah P 03/22/2026 0 99999 7 -1  
aira@aira-pc:~$ sudo passwd -S mike  
mike P 03/22/2026 0 99999 7 -1
```

Verify Created passwords of sarah nad mike

sudo passwd -S sarah

sudo passwd -S mike

```

aira@aira-pc:~$ echo "sarah:SecurePass123!" | sudo chpasswd
aira@aira-pc:~$ echo "mike:SecurePass456!" | sudo chpasswd
aira@aira-pc:~$ sudo passwd -S sarah
sarah P 03/22/2026 0 99999 7 -1
aira@aira-pc:~$ sudo passwd -S mike
mike P 03/22/2026 0 99999 7 -1

```

Create and secure directories for Sarah and mike

`sudo mkdir -p /home/sarah/workspace /home/mike/workspace`

`ls -ld /home/sarah/workspace /home/mike/workspace`

```

aira@aira-pc:~$ sudo mkdir -p /home/sarah/workspace /home/mike/workspace
aira@aira-pc:~$
aira@aira-pc:~$
aira@aira-pc:~$
aira@aira-pc:~$ ls -ld /home/sarah/workspace /home/mike/workspace
drwxr-xr-x 2 root root 4096 Mar 22 12:04 /home/mike/workspace
drwxr-xr-x 2 root root 4096 Mar 22 12:04 /home/sarah/workspace

```

Granting User Ownership of Home Subdirectories

`sudo chown sarah:sarah /home/sarah/workspace`

`sudo chown mike:mike /home/mike/workspace`

```

aira@aira-pc:~$ sudo chown sarah:sarah /home/sarah/workspace
/home/mike/workspaceaira@aira-pc:~$ sudo chown mike:mike /home/mike/workspace
aira@aira-pc:~$

```

Isolating Sarah and mikes's Workspace Directories

`sudo chmod 700 /home/sarah/workspace /home/mike/workspace`

```

aira@aira-pc:~$ sudo chmod 700 /home/sarah/workspace /home/mike/workspace
aira@aira-pc:~$

```

Implement Password Policy

Set password expiration to 30 days

`sudo chage -M 30 sarah`

`sudo chage -M 30 mike`

Verify 30 days Password created for Sarah and mike

`sudo chage -l sarah`

`sudo chage -l mike`

```

aira@aira-pc:~$ sudo chage -l sarah
Last password change           : Mar 22, 2026
Password expires                : Apr 21, 2026
Password inactive               : never
Account expires                 : never
Minimum number of days between password change : 0
Maximum number of days between password change : 30
Number of days of warning before password expires : 7
aira@aira-pc:~$ sudo chage -l mike
Last password change           : Mar 22, 2026
Password expires                : Apr 21, 2026
Password inactive               : never
Account expires                 : never
Minimum number of days between password change : 0
Maximum number of days between password change : 30

```

Implementing a Strong Password Complexity Policy

To enforce complexity (on Debian/Ubuntu), edit `/etc/pam.d/common-password`:

Add:

```
password requisite pam_pwquality.so retry=3 minlen=12 lcredit=-1 ucredit=-1  
dcredit=-1 ocredit=-1
```

```
# here are the per-package modules (the "Primary" block)  
password requisite pam_pwquality.so retry=3 minlen=12 lcredit=-1 ucredit=-1 dcredit=-1 ocredit=-1  
password [success=1 default=ignore] pam_unix.so obscure sha512  
# here's the fallback if no module succeeds  
password requisite pam_deny.so  
# prime the stack with a positive return value if there isn't one already;  
# this avoids us returning an error just because nothing sets a success code
```

How to Verify

It should now give you an error if the password doesn't meet your 12-character, multi-case and symbol requirements

```
sudo passwd sarah
```

Task 3:

```
aira@aira-pc:~$ sudo passwd sarah  
New password:  
BAD PASSWORD: The password is shorter than 8 characters  
Retype new password:  
Sorry, passwords do not match.  
New password:  
BAD PASSWORD: The password contains the user name in some form
```

Backup Configuration for Web Servers

Create Backup Directory:

```
sudo mkdir -p /backups  
sudo chmod 777 /backups
```

```
aira@aira-pc:~$ ls -la /backups/  
total 8  
drwxrwxrwx  2 root root 4096 Mar 22 12:39   
drwxr-xr-x 20 root root 4096 Mar 22 12:39 ..  
aira@aira-pc:~$
```

Sarah's Backup Script (`/home/sarah/backup.sh`)

```
cd /home/sarah/
```

```
make backup.sh
```

```
#!/bin/bash
```

```
DATE=$(date +%Y-%m-%d)
```

```
FILE="/backups/apache_backup_${DATE}.tar.gz"
```

```
tar -czf $FILE /etc/httpd/ /var/www/html/
```

```
echo "Verification for $DATE:" >> /backups/verify.log
```

```
tar -tf $FILE >> /backups/verify.log
```

```
aira@aira-pc:/home/sarah$ vi backup.sh  
aira@aira-pc:/home/sarah$ sudo vi backup.sh  
aira@aira-pc:/home/sarah$  
aira@aira-pc:/home/sarah$ sudo vi backup.sh  
aira@aira-pc:/home/sarah$
```

Mike's Backup Script (/home/mike/backup.sh)

```
cd /home/mike/  
make backup.sh
```

```
#!/bin/bash  
DATE=$(date +%Y-%m-%d)  
FILE="/backups/nginx_backup_$DATE.tar.gz"  
tar -czf $FILE /etc/nginx/ /usr/share/nginx/html/  
echo "Verification for $DATE:" >> /backups/verify.log  
tar -tf $FILE >> /backups/verify.log
```

```
aira@aira-pc:/home/mike$  
aira@aira-pc:/home/mike$ cd /home/mike/  
aira@aira-pc:/home/mike$ sudo vim backup.sh  
aira@aira-pc:/home/mike$
```

Granting Execution Rights to User Scripts

```
sudo chown sarah:sarah /home/sarah/backup.sh  
sudo chown mike:mike /home/mike/backup.sh
```

```
aira@aira-pc:/home/mike$ sudo chmod +x /home/sarah/backup.sh /home/mike/backup.sh  
aira@aira-pc:/home/mike$ ls -la /home/sarah/backup.sh  
-rwxr-xr-x 1 root root 211 Mar 22 12:42 /home/sarah/backup.sh  
aira@aira-pc:/home/mike$ ls -la /home/mike/backup.sh  
-rwxr-xr-x 1 root root 218 Mar 22 12:51 /home/mike/backup.sh  
aira@aira-pc:/home/mike$
```

Automating Weekly Backup Schedules via Cron jobs

This job has to run every Tuesday at 12:00 AM

```
sudo crontab -e
```

```
0 0 * * 2 /bin/bash /home/sarah/backup.sh  
0 0 * * 2 /bin/bash /home/mike/backup.sh
```

```
aira@aira-pc: /home/mike  
# Edit this file to introduce tasks to be run by cron.  
#  
# Each task to run has to be defined through a single line  
# indicating with different fields when the task will be run  
# and what command to run for the task  
#  
# To define the time you can provide concrete values for  
# minute (m), hour (h), day of month (dom), month (mon),  
# and day of week (dow) or use '*' in these fields (for 'any').  
#  
# Notice that tasks will be started based on the cron's system  
# daemon's notion of time and timezones.  
#  
# Output of the crontab jobs (including errors) is sent through  
# email to the user the crontab file belongs to (unless redirected).  
#  
# For example, you can run a backup of all your user accounts  
# at 5 a.m every week with:  
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/  
#  
# For more information see the manual pages of crontab(5) and cron(8)  
#
```

The Breakdown of 0 0 * * 2

Cron uses five fields to determine the timing. Here is how yours is set up:

Field	Value	Description
Minute	0	At the start of the hour (00)
Hour	0	At midnight (00:00)
Day of Month	*	Every day of the month
Month	*	Every month
Day of Week	2	Tuesday (0=Sun, 1=Mon, 2=Tue...)

How to verify it is scheduled

Run this command to list all active cron jobs and make sure yours are there

```
sudo crontab -l
```

```
sarah@sarah-pc:/home/mike$ sudo crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
0 0 * * 2 /bin/bash /home/sarah/backup.sh
0 0 * * 2 /bin/bash /home/mike/backup.sh
```

Check the Cron Service Status

```
sudo systemctl status cron
```

```

aira@aira-pc:/home/mike$ sudo systemctl status cron
● cron.service - Regular background program processing daemon
   Loaded: loaded (/lib/systemd/system/cron.service; enabled; vendor preset: enabled)
   Active: active (running) since Sun 2026-03-22 11:37:39 IST; 1h 51min ago
     Docs: man:cron(8)
  Main PID: 351 (cron)
    Tasks: 1 (limit: 4509)
   Memory: 584.0K
      CPU: 71ms
   CGroup: /system.slice/cron.service
           └─351 /usr/sbin/cron -f

Mar 22 11:37:39 aira-pc systemd[1]: Started Regular background program processing daemon.
Mar 22 11:37:39 aira-pc cron[351]: (CRON) INFO (pidfile fd = 3)
Mar 22 11:37:39 aira-pc cron[351]: (CRON) INFO (Running @reboot jobs)
Mar 22 12:17:01 aira-pc CRON[1742]: pam_unix(cron:session): session opened for user root by (uid=0)
Mar 22 12:17:01 aira-pc CRON[1743]: (root) CMD ( cd / && run-parts --report /etc/cron.hourly)
Mar 22 12:17:01 aira-pc CRON[1742]: pam_unix(cron:session): session closed for user root
Mar 22 13:17:01 aira-pc CRON[3008]: pam_unix(cron:session): session opened for user root by (uid=0)
Mar 22 13:17:01 aira-pc CRON[3008]: pam_unix(cron:session): session closed for user root
aira@aira-pc:/home/mike$

```

Run for Verification.

To confirm everything is working *now* without waiting until Tuesday, you can temporarily set a job to run every minute:

Steps:

- Open crontab: `sudo crontab -e`
- Add: `***** /bin/bash /home/sarah/backup.sh`
`***** /bin/bash /home/mike/backup.sh`
- Save and wait 60 seconds, then check the logs again using the `grep` commands above.

```
aira@aira-pc: /home/mike
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow  command
* * * * * /bin/bash /home/sarah/backup.sh
* * * * * /bin/bash /home/mike/backup.sh

"/tmp/crontab.Iu1M5q/crontab" 26L, 973C
```

Backups for Every 60 Seconds:

ls -lrt /backups/

```
aira@aira-pc:/home/mike$ ls -lrt /backups/
total 12
-rw-r--r-- 1 root root 348 Mar 22 13:57 verify.log
-rw-r--r-- 1 root root 45 Mar 22 13:57 nginx_backup_2026-03-22.tar.gz
-rw-r--r-- 1 root root 45 Mar 22 13:57 apache_backup_2026-03-22.tar.gz
aira@aira-pc:/home/mike$
```

Implementation Report Summary:

Task 1: Tools like `htop` provide real-time visibility, while the custom script ensures historical data for capacity planning.

Task 2: Isolation was achieved using `chmod 700`, preventing users from "peeking" into each other's workspaces. Password aging was enforced via `chage`.

Task 3: Automation via `cron` ensures backups occur without manual intervention. The inclusion of `tar -tz` in the script provides an automated integrity log.

Challenges: Ensuring the `/backups` directory had correct permissions for both user scripts to write to was critical. For production, consider moving backups to an external volume.